Engine Model(s): PT6A

72-20-00

AIR INLET SECTION - DESCRIPTION AND OPERATION

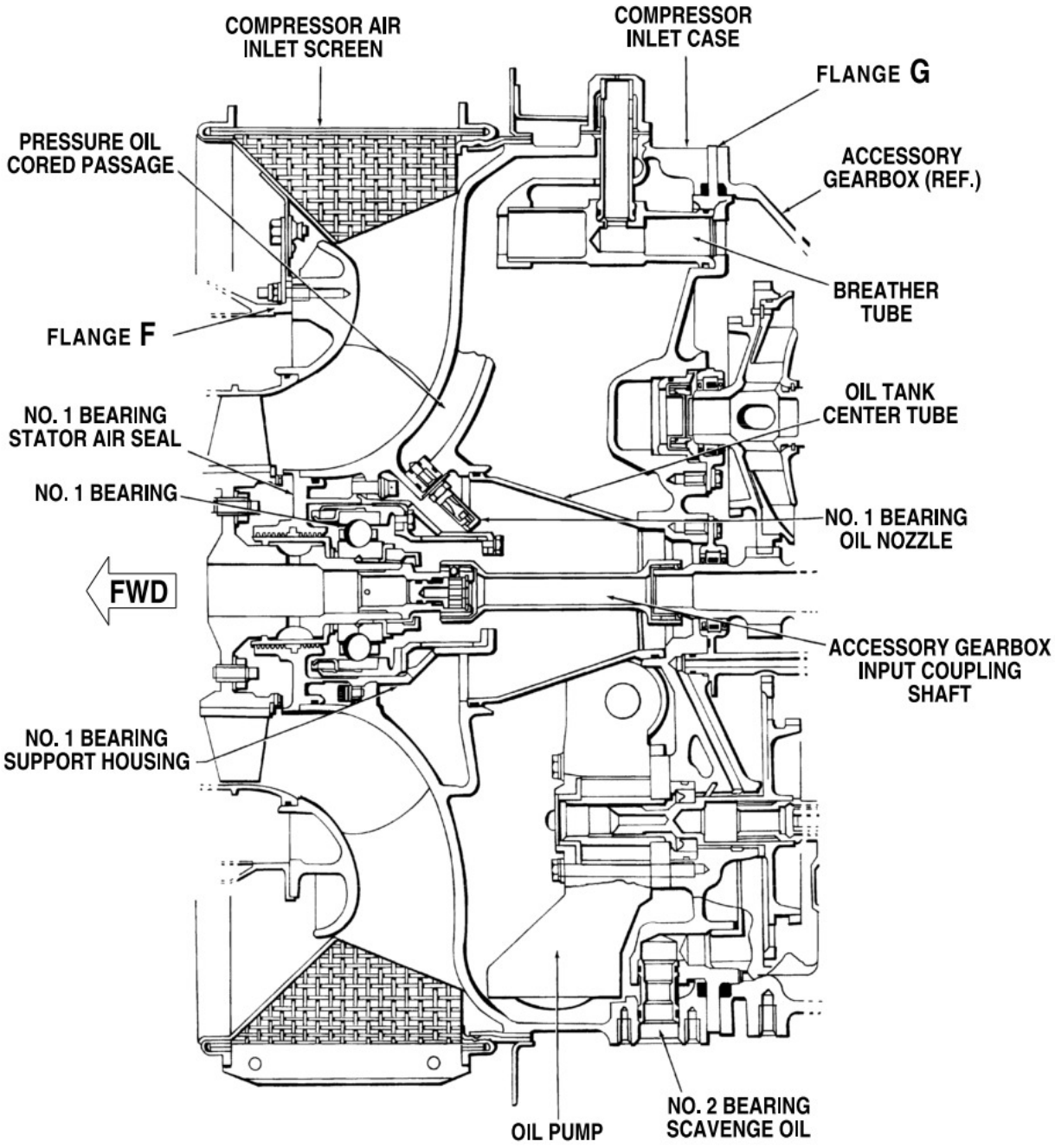
1. General

A. The compressor inlet case is a component of the engine gas generator assembly and directs incoming air to the compressor rotor assembly. An air inlet screen prevents debris entering the compressor inlet.

2. Description and Operation (Ref. Fig. 1)

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Figure 1 Compressor Inlet Case and Air Inlet Screen



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A. Compressor Inlet Case

The compressor inlet case consists of a circular light alloy casting; the front section of which forms an annular plenum chamber for the passage of the inlet air to the compressor. The rear section is a hollow compartment that is used to form the front section of the integral oil tank. Two mounting flanges are incorporated; flange F, at the front end of the inlet area, provides the mounting for the gas generator case, while flange G, at the rear of the oil tank area, provides the mounting flange for the accessory gearbox. A circular wire mesh screen is bolted around the air inlet area of the case to prevent foreign object ingestion by the compressor.

The No. 1 bearing, bearing support housings and compressor rear stator air seal are mounted in the centerbore of the compressor inlet case.The bearing outer housing and rear stator air seal are secured within the flange by four bolts. The No. 1 bearing outer race is secured within the bearing inner housing by an externally threaded retaining ring and locking rivet, while the inner housing is secured to the outer housing by four bolts and nuts at the rear.

A calibrated oil nozzle, installed at the end of a cored passage in the compressor inlet case at the approximate 7 o'clock position, provides lubrication for the No. 1 bearing. A cored outlet port at the 6 o'clock position allows scavenge oil to flow to the accessory gearbox diaphragm.

Compressor interstage air is applied to the center of the rear stator air seal to prevent oil from the No. 1 bearing area flowing into the compressor section. Part of the air, at the center of the seal, vents forward and into the inlet air stream while the remainder vents to the rear and into the accessory gearbox via the oil tank center tube.

The oil filter, oil filter housing, check and bypass valve assembly (Ref. 79-20-04) is installed at the 3 o'clock position on the inlet case with the check valve located in the outlet port of the oil pump (Ref. 72-60-00). A cover plate, with teflon spacer, bolted to the inlet case retains the oil filter and housing. A tube, with preformed packings, is installed between the centerbore of the inlet case and the accessory gearbox diaphragm to provide a passage for the accessor drive coupling shaft (Ref. 72-60-00).

B. Air Inlet Screen

The air inlet screen is a circular wire mesh screen that is bolted around the air inlet area of the inlet case to prevent foreign object ingestion by the compressor.

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